

9 Manifolds|Fenichel's Theorem

(Recall) Singular Perturbation Systems: $0 < \varepsilon \ll 1$
fast $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \dot{v} = \varepsilon g(u, v, \varepsilon) \end{cases} \xLeftrightarrow{\tau = \varepsilon t}$ slow $\begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \xrightarrow{\varepsilon = 0}$ layer $\begin{cases} \dot{u} = f(u, v, 0) \\ \dot{v} = 0 \end{cases}$ reduced $\begin{cases} 0 = f(u, v, 0) \\ v' = g(u, v, 0) \end{cases}$
ps: fast var. u (can be vector); slow var. v .
Critical Manifold: $M_0 := \{(u, v) : f(u, v, 0) = 0\}$ ps: reduced problem is defined on M_0
Normal Hyperbolicity: Critical manifold M_0 is normally hyperbolic if:
[$\Re(\lambda_i) \neq 0$ (特征值) of $J = [f_{u_1} \ f_{u_2} \ \dots]$] $\Rightarrow$ For u 1-dim: If $\frac{\partial f}{\partial u} \neq 0$
If $\Re(\lambda_i) < 0 (> 0) \Rightarrow \frac{\partial f}{\partial u} < 0 (> 0)$: M_0 is **normally attracting** (stable) (**normally repelling** (unstable)).
Fenichel's Theorem: If $M_0 \subseteq \{(u, v) : f(u, v, 0) = 0\}$ is compact (closed & bounded) and normally hyperbolic.
Suppose f, g are smooth. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ Manifold M_ε s.t. it's $\mathcal{O}(\varepsilon)$ close to M_0 and diffeomorphic to M_0 . (i.e. M_ε is locally invariant under the flow of the full system.) M_ε called **slow manifold**.
Find Slow Manifold M_ε : If $M_0 = \{(u, v) : u = h_0(v)\}$, then $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)\}$
e.g. Given exact sol: $(u, v) = ((u_0 - \frac{v_0^2}{1-2\varepsilon})e^{-t} + \frac{v_0^2}{1-2\varepsilon}e^{-2\varepsilon t}, v_0e^{-\varepsilon t})$. Given $M_0 = \{(u, v) : u = v^2\}$.
Then 1. Linearise, $\tau = \varepsilon t \Rightarrow (u, v) = (\frac{v_0^2}{1-2\varepsilon}e^{-2\tau}, v_0e^{-\tau}) \Rightarrow u = \frac{v^2}{1-2\varepsilon}$. 2. Check, when $\varepsilon = 0, u = v^2$.
3. Therefore, $M_\varepsilon = \{(u, v) : u = \frac{v^2}{1-2\varepsilon}\}$. If $[u]' = [\frac{v^2}{1-2\varepsilon}]'$, it's **invariant**. (under slow flow)
Invariance of Manifold: $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)\}$, if $[u]' = [h_\varepsilon(v)]'$, $' = d/d\tau$ 或对应的.

10 Havesting Model

Discrete Population Model: $N_{t+1} = N_t + rN_t(1 - \frac{N_t}{K}) - H_t$, ps: 分析类似 continuous model.
- If $N_{\min} < 1$ **species eradicated/extinct**.
- **Recovery Time**: $T = -\frac{1}{\lambda}$, where $\lambda = f'(N^*)$
Continuous Population Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - h(N)$, r : growth rate, K : carrying capacity.
 $h(N)$: harvesting term. a.constant: $h(N) = Y_0$; b.density-dependent: $h(N) = EN$ (E : harvesting effort).
For the density-dependent harvesting, we have:
- If $E > r$, species eradicated/extinct. If $E < r$, we have the following:
- **Maximum (Sustainable) Yield**: $= \max_N h(N) = h(\frac{K}{2}) = \frac{rK}{4}$ ps: at $N^* = \frac{K}{2}, E = \frac{r}{2}$
Improved our Model: Combine both harvesting types: $h(N) = EN$ for N small, $h(N) = Y_0$ for N large.

coalesce : 两点合并	eradicated: 根除	membrane potential: 膜电位	unimodal: 单峰	vaccination: 疫苗接种
predator: 捕食者	prey: 被捕食者	trnsriptase: 反转录酶	invades: 入侵	plausible: 似乎合理的
invasion: 入侵	intraspecific competition: 种内竞争		interspecific competition: 种间竞争	
promote: 促进	suppress: 抑制	eliminate: 消除	repelling: 排斥的	
boundary region = trapping region = invariant region = confined region				

Taylor : $\sin(x) \simeq x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	$\cos(x) \simeq 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	$\tan(x) \simeq x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$
$\sinh(x) \simeq x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$	$\cosh(x) \simeq 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$	$\tanh(x) \simeq x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$
Hyperbolic : $\cosh^2(x) - \sinh^2(x) = 1$	$1 + \tanh^2(x) = \sec^2(x)$	$\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$
Integral : $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(\frac{x}{a})$	$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \operatorname{arcosh}(\frac{x}{a})$	$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan(\frac{x}{a})$
$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \operatorname{arcsinh}(\frac{x}{a})$	$\int \frac{1}{x^2 - a^2} dx = \frac{1}{a} \operatorname{arctanh}(\frac{x}{a})$	

11 Regular/Singular Perturbation Equation

Singular: If eq like: $\varepsilon y'' + a(x, \varepsilon)y' + b(x, \varepsilon)y = 0$ (i.e. highest der. mult. by ε)
Regular: If eq like: $y'' + a(x, \varepsilon)\varepsilon y' + b(x, \varepsilon)y = 0$ (i.e. highest der. NOT mult. by ε)
Approximation Solution: Assume sol: $y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$
Substitute into ODE, compare coeff. of $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc. 不留 $Y_0 = 0$ 解, 前面已经得到过
- If **singular**, we cannot approximate the other root. $\rightarrow Y = \varepsilon y$, and get new ODE. $\downarrow$
- Assume sol: $Y(x; \varepsilon) = Y_0(x) + \varepsilon Y_1(x) + \varepsilon^2 Y_2(x) + \dots$, substitute, compare $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc.
***Uniformly Valid Approximation/Expression**: sol = outer sol + inner sol - matching part
where matching part = section.7 Asymptotic Matching 的结果

12 Appendix

Desingularisation: For $\begin{cases} U' = f(U, V) \\ V' = \frac{g(U, V)}{U} \end{cases}$ or $X' = \frac{F(X)}{h(X)} \Rightarrow$ Introduce new variable τ s.t.
 $\frac{dt}{d\tau} = U$ or $\frac{dt}{d\tau} = h(X)$ Then $\begin{cases} \frac{dU}{d\tau} = U f(U, V) \\ \frac{dV}{d\tau} = g(U, V) \end{cases}$ or $\frac{dX}{d\tau} = F(X)$
对于证明 $V(U)$ 是 **heteroclinic connection**: 正常证明 + $V(U) \rightarrow V^\pm$ as $U \rightarrow U^\pm$ So that It is a heteroclinic connection from the unstable manifold $W^u(P^-)$ to the stable manifold $W^s(P^+)$. ($P^\pm$ 是 equilibrium point)